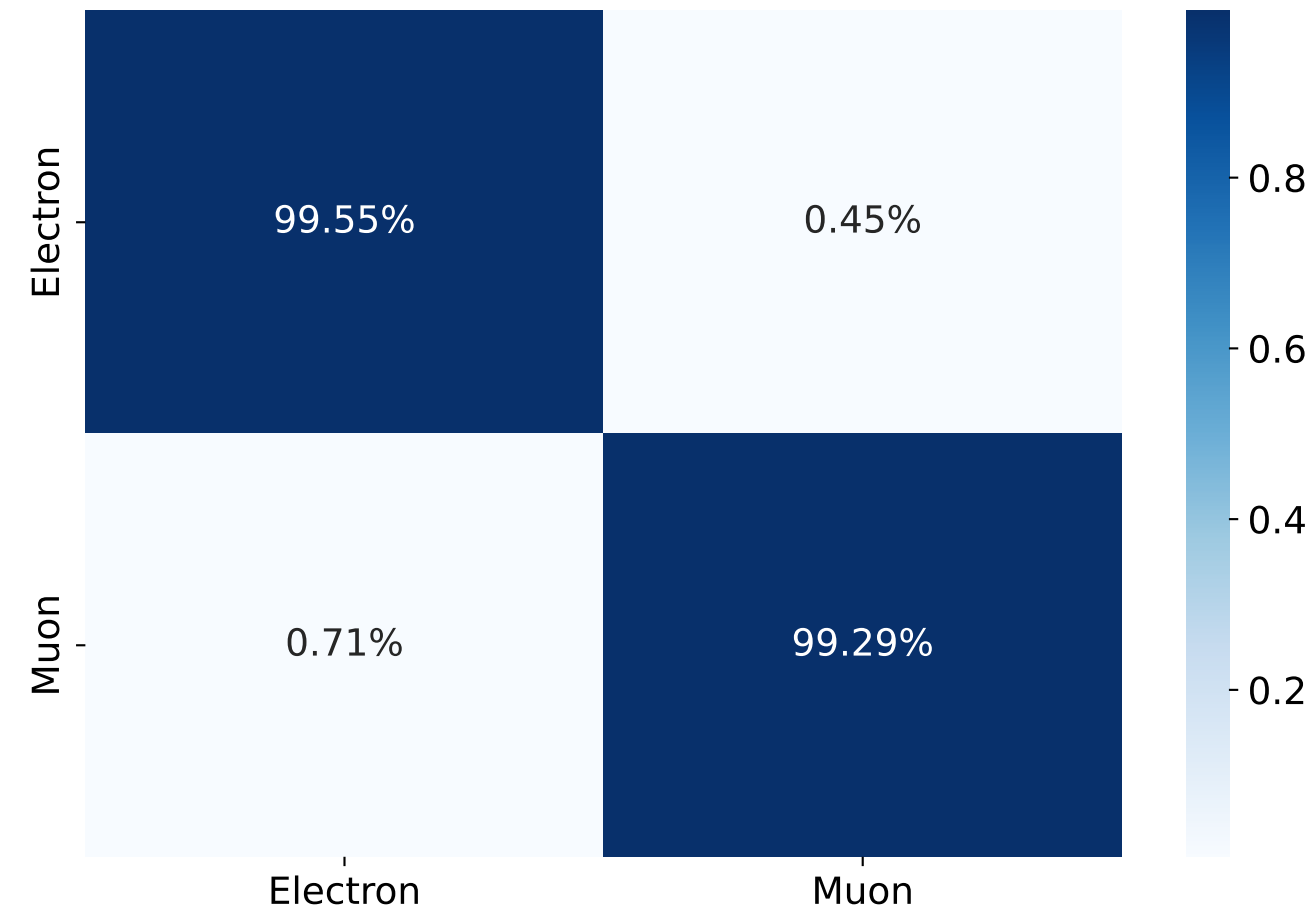
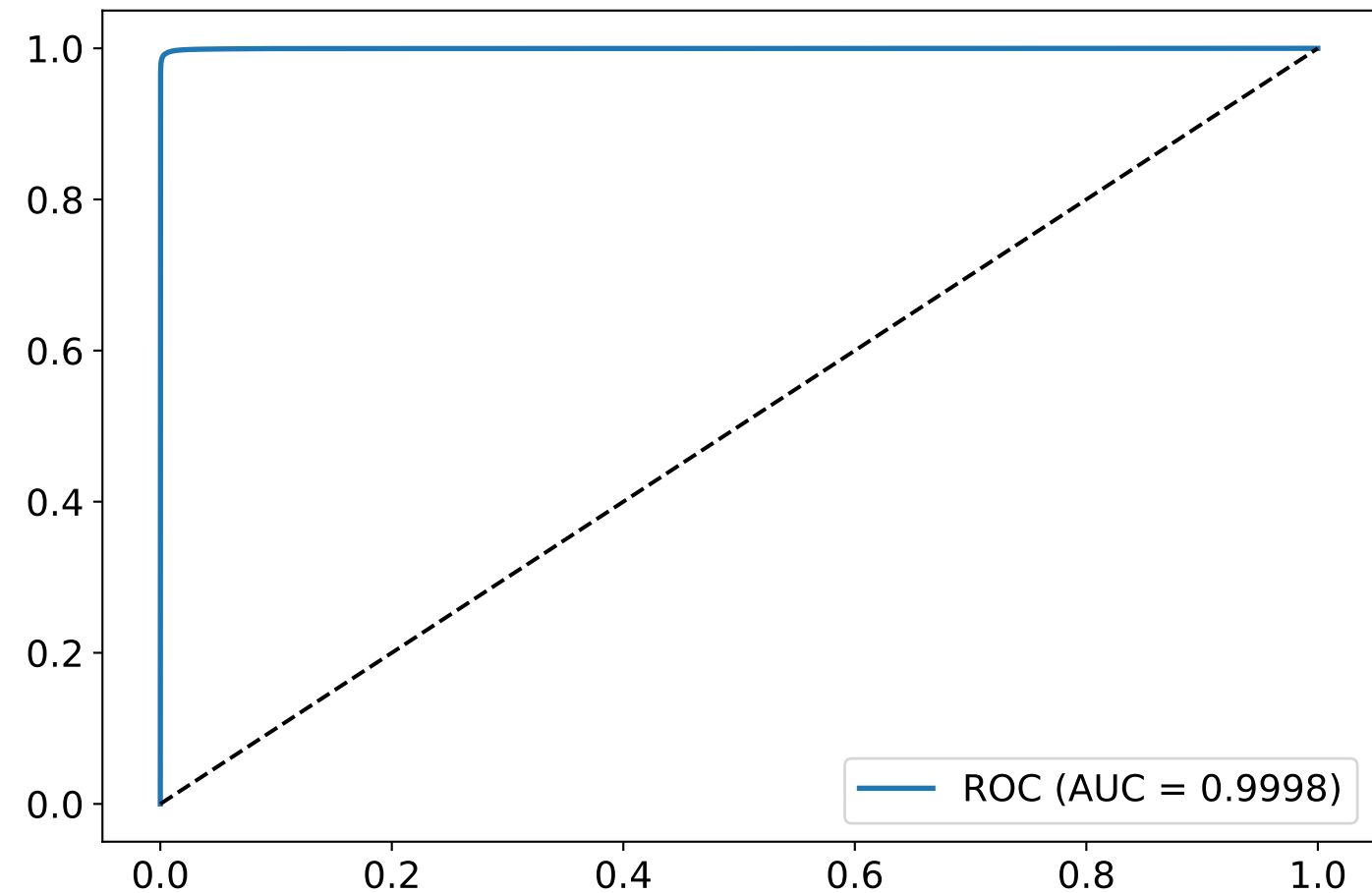


EtaRange = FullRange (TEST DATA) - Accuracy: 0.994

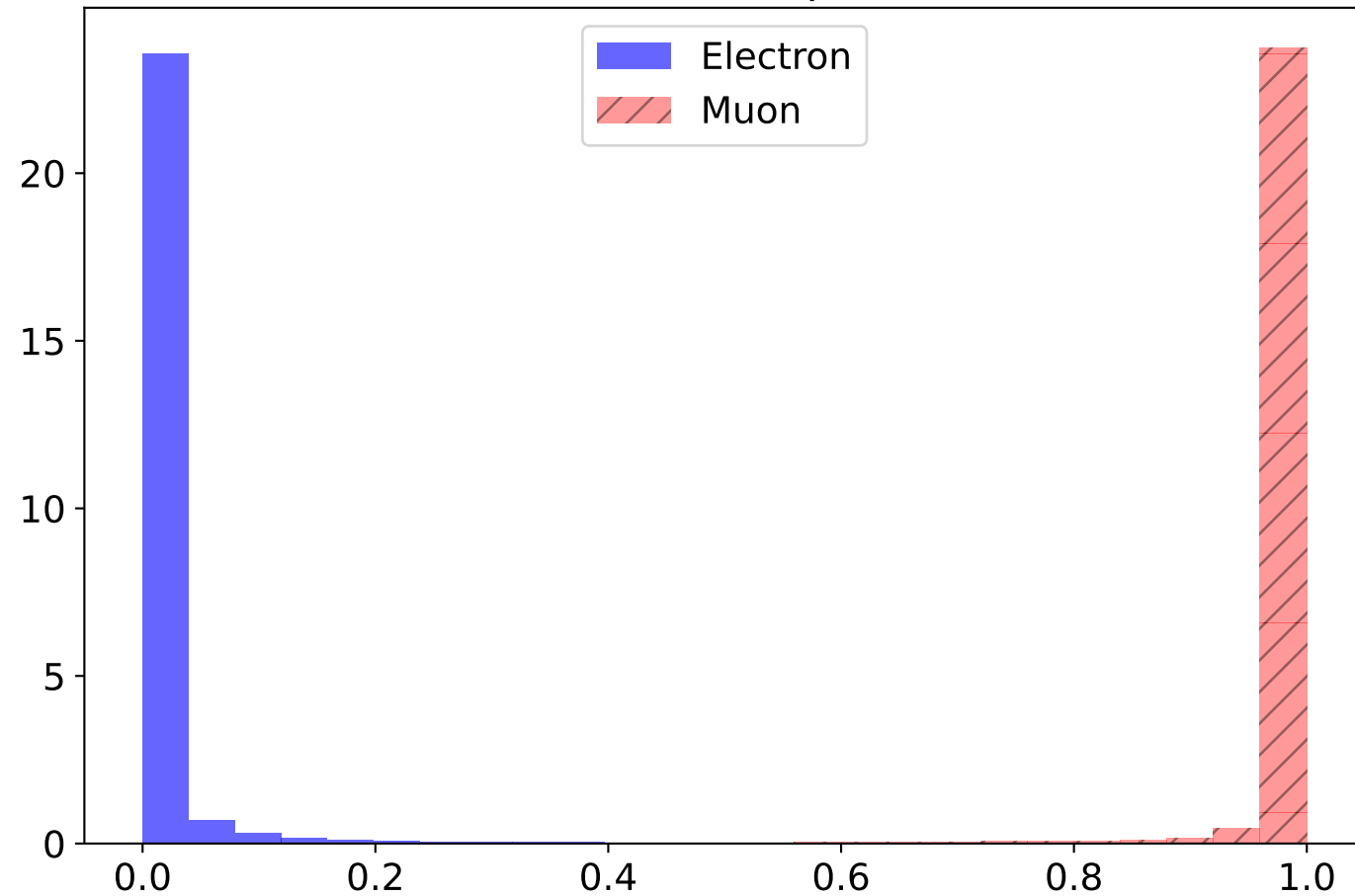
Confusion Matrix



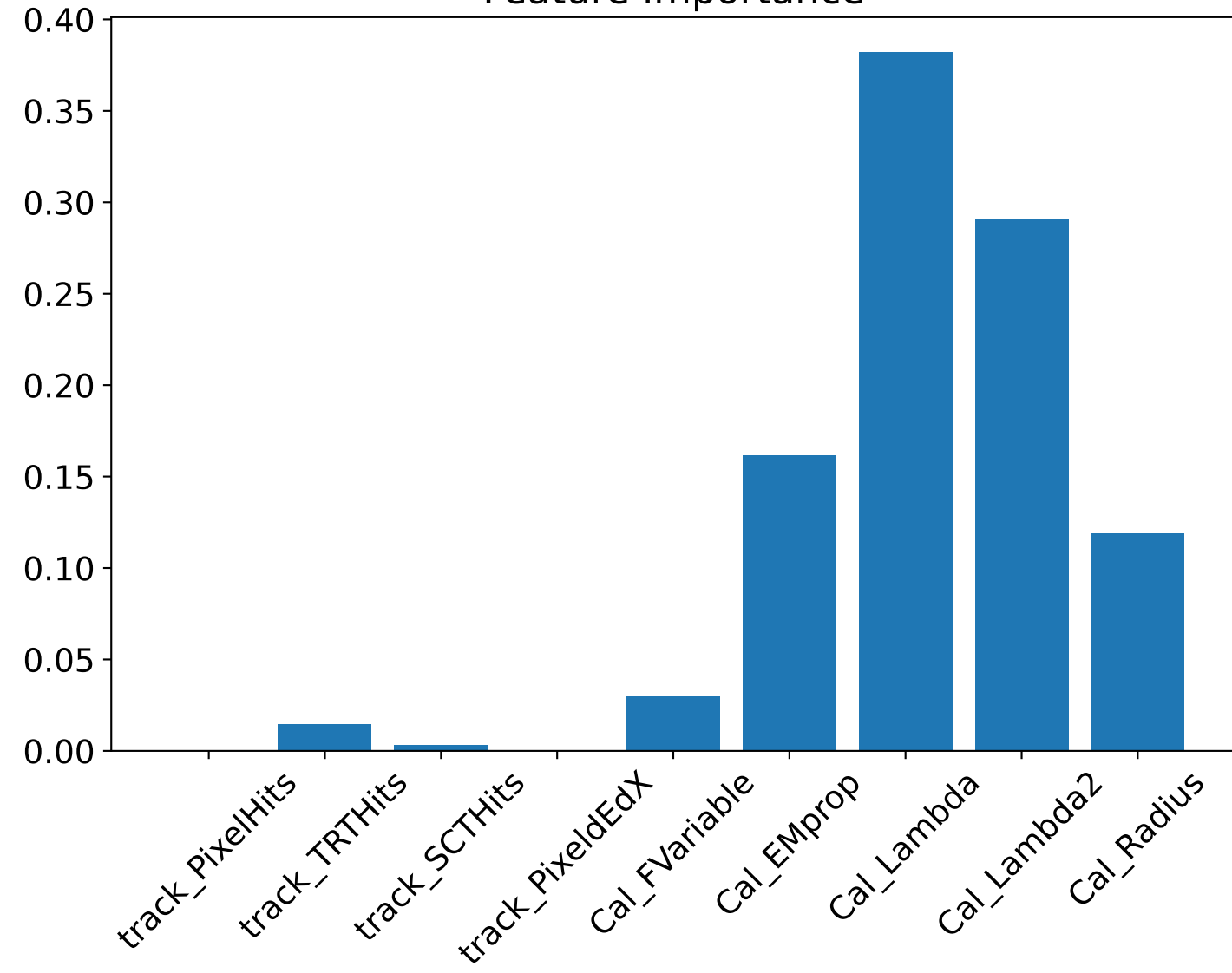
ROC Curve



Classifier Response

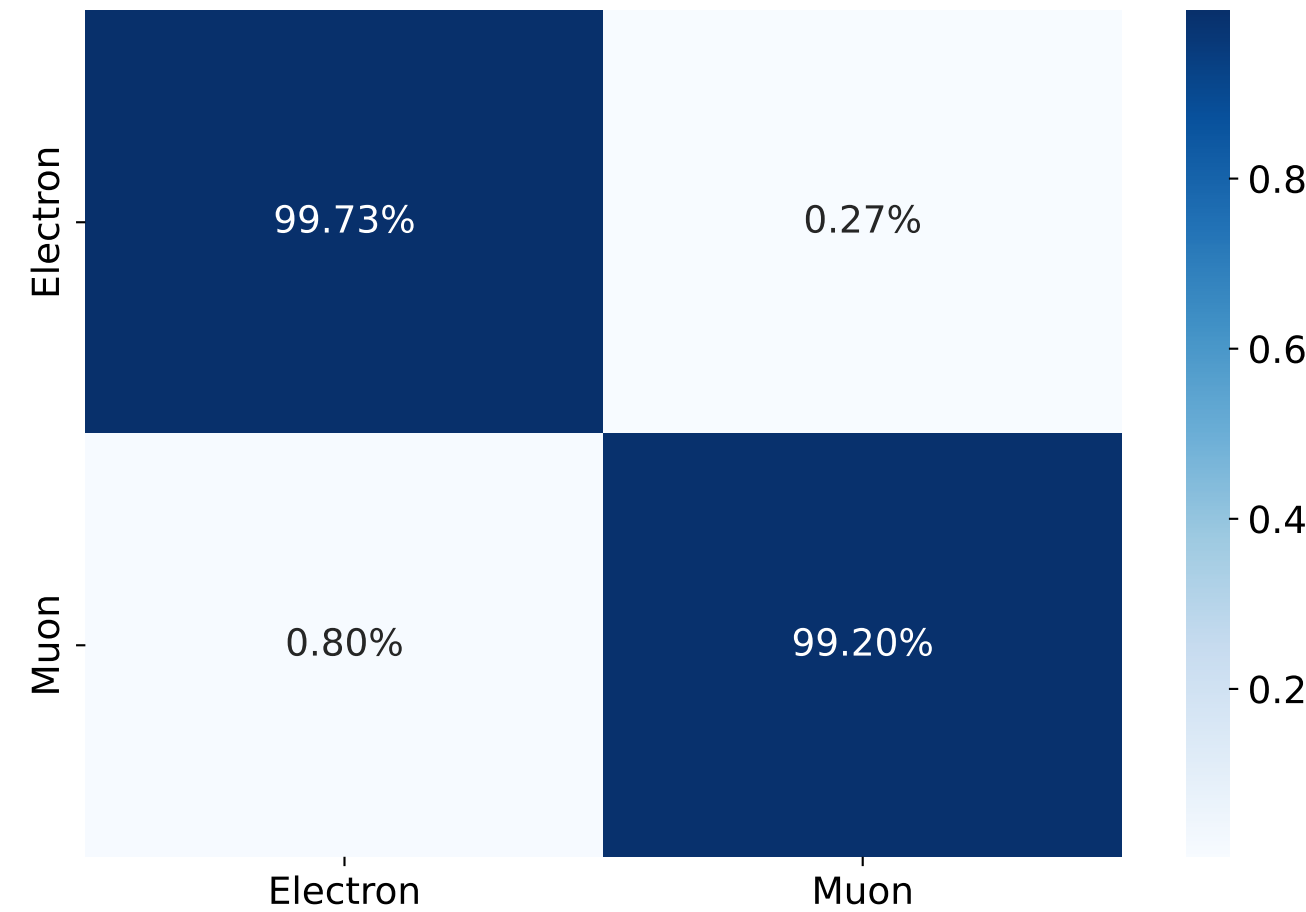


Feature Importance

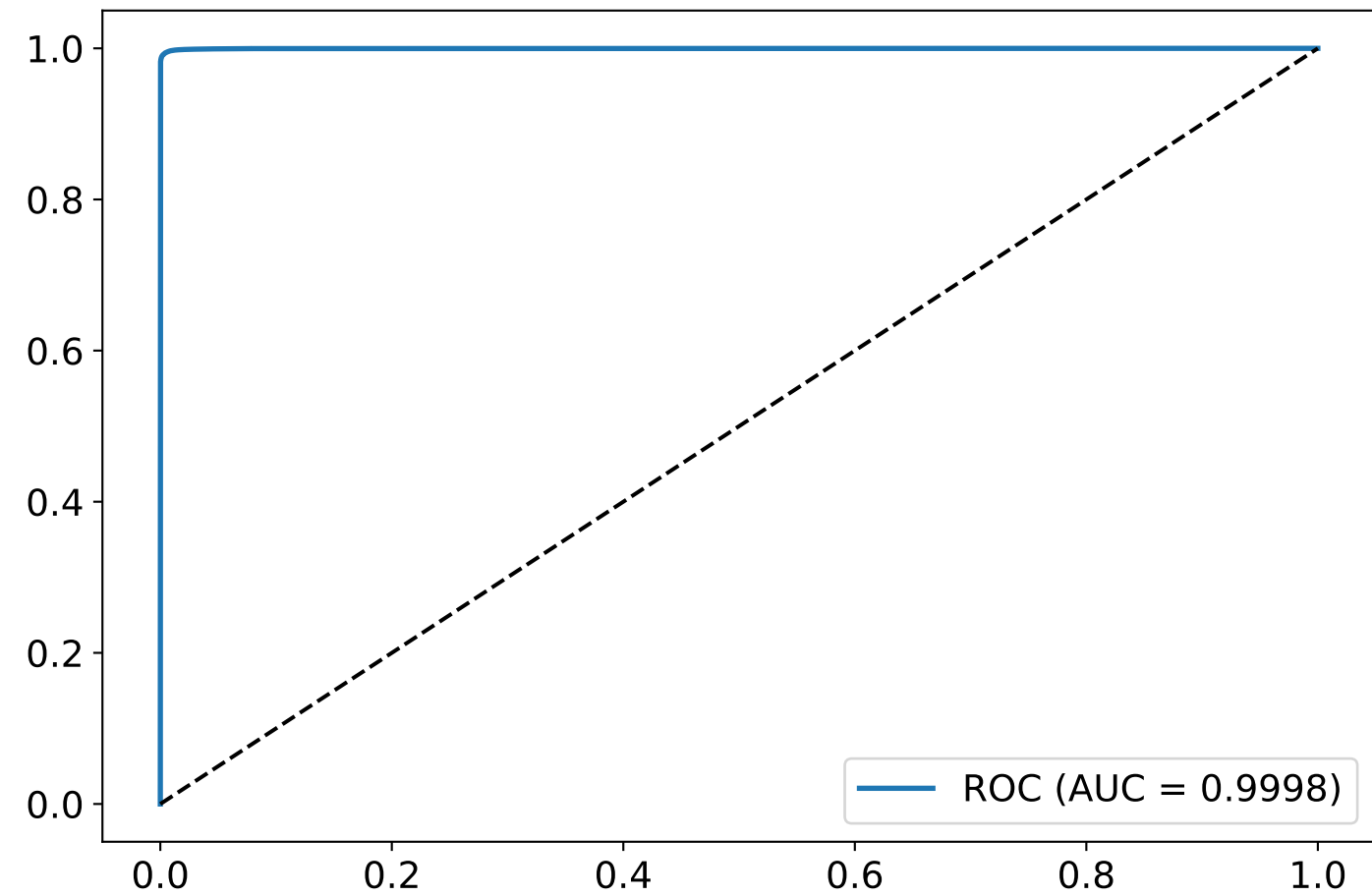


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.994

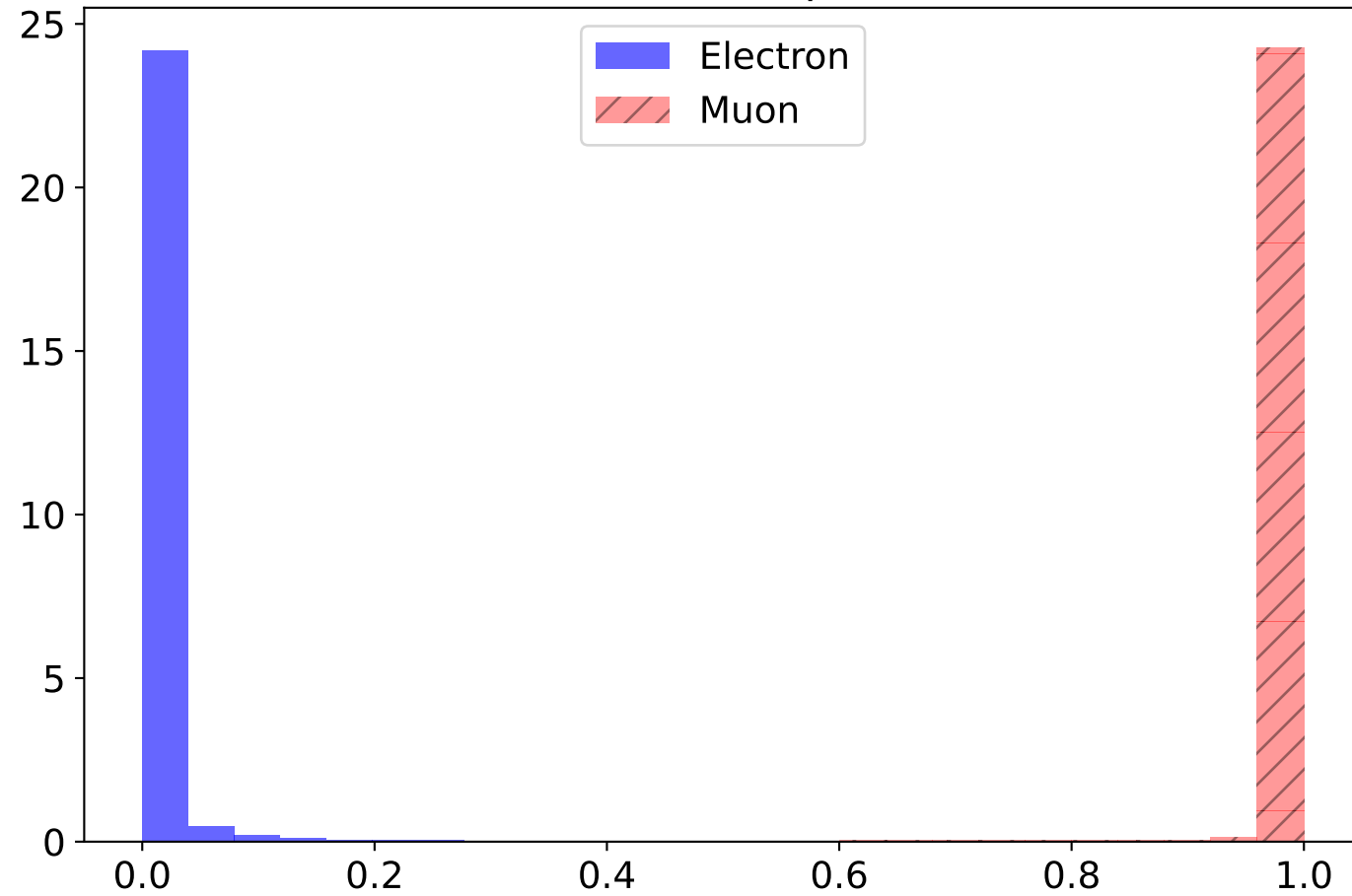
Confusion Matrix



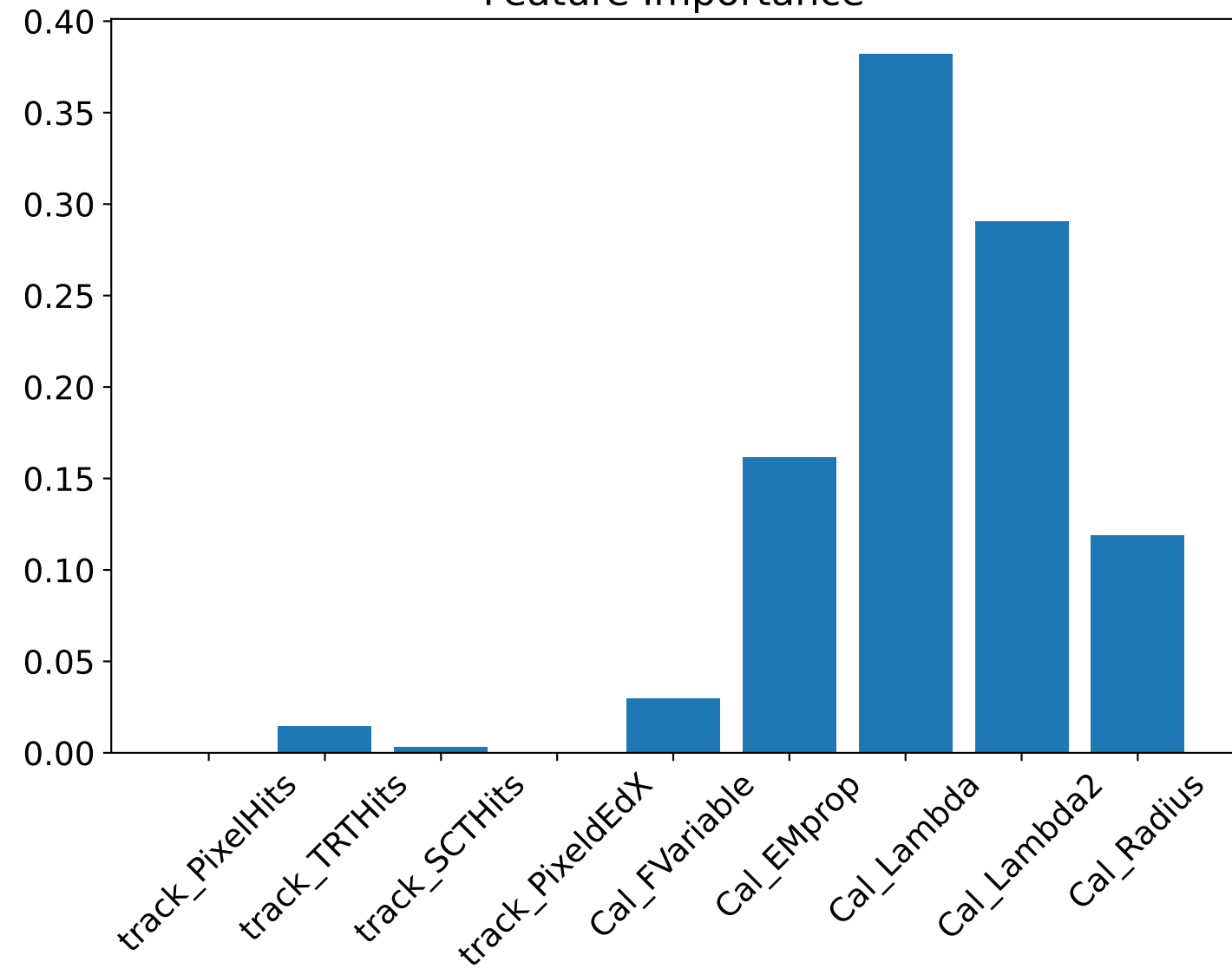
ROC Curve



Classifier Response

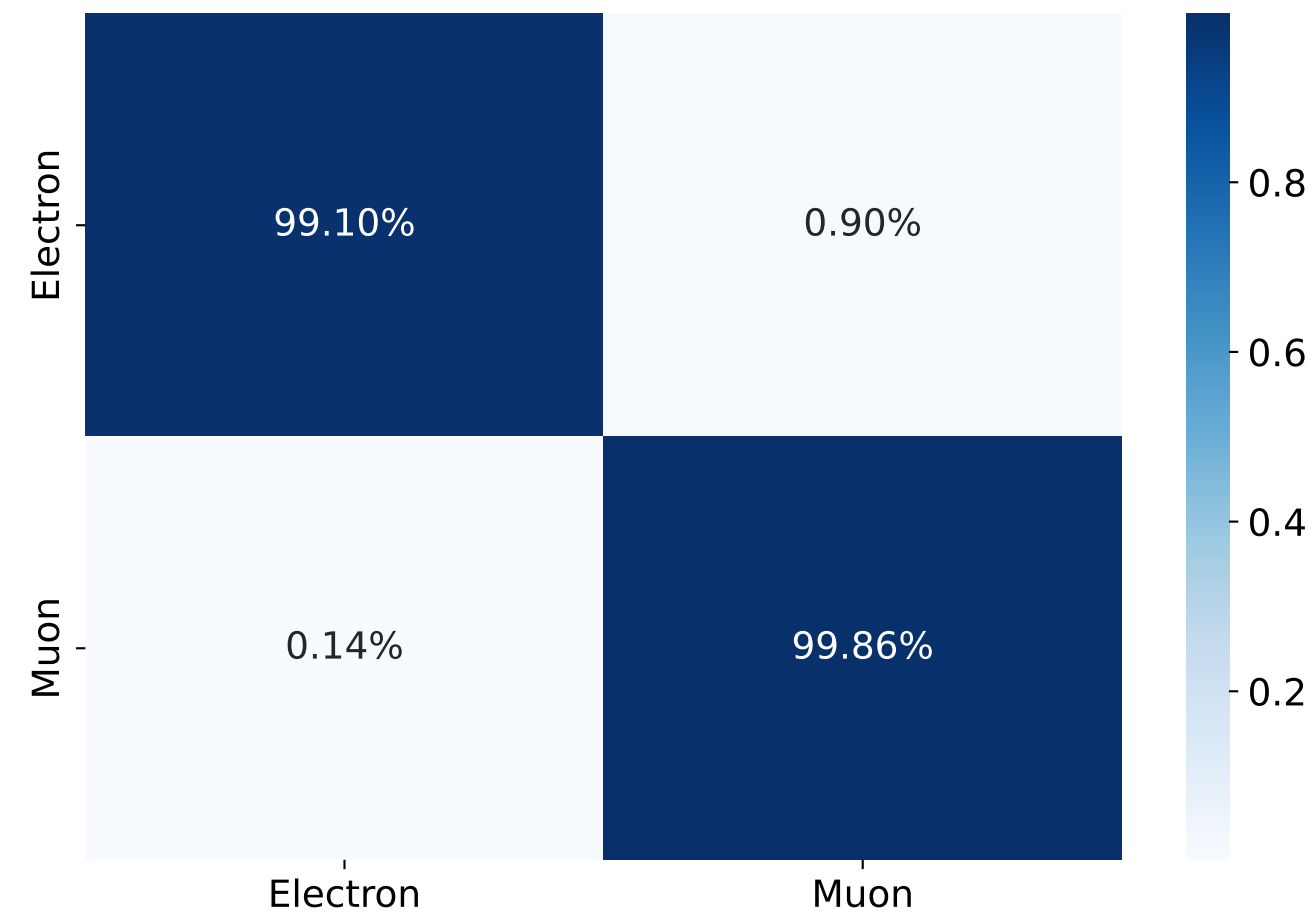


Feature Importance

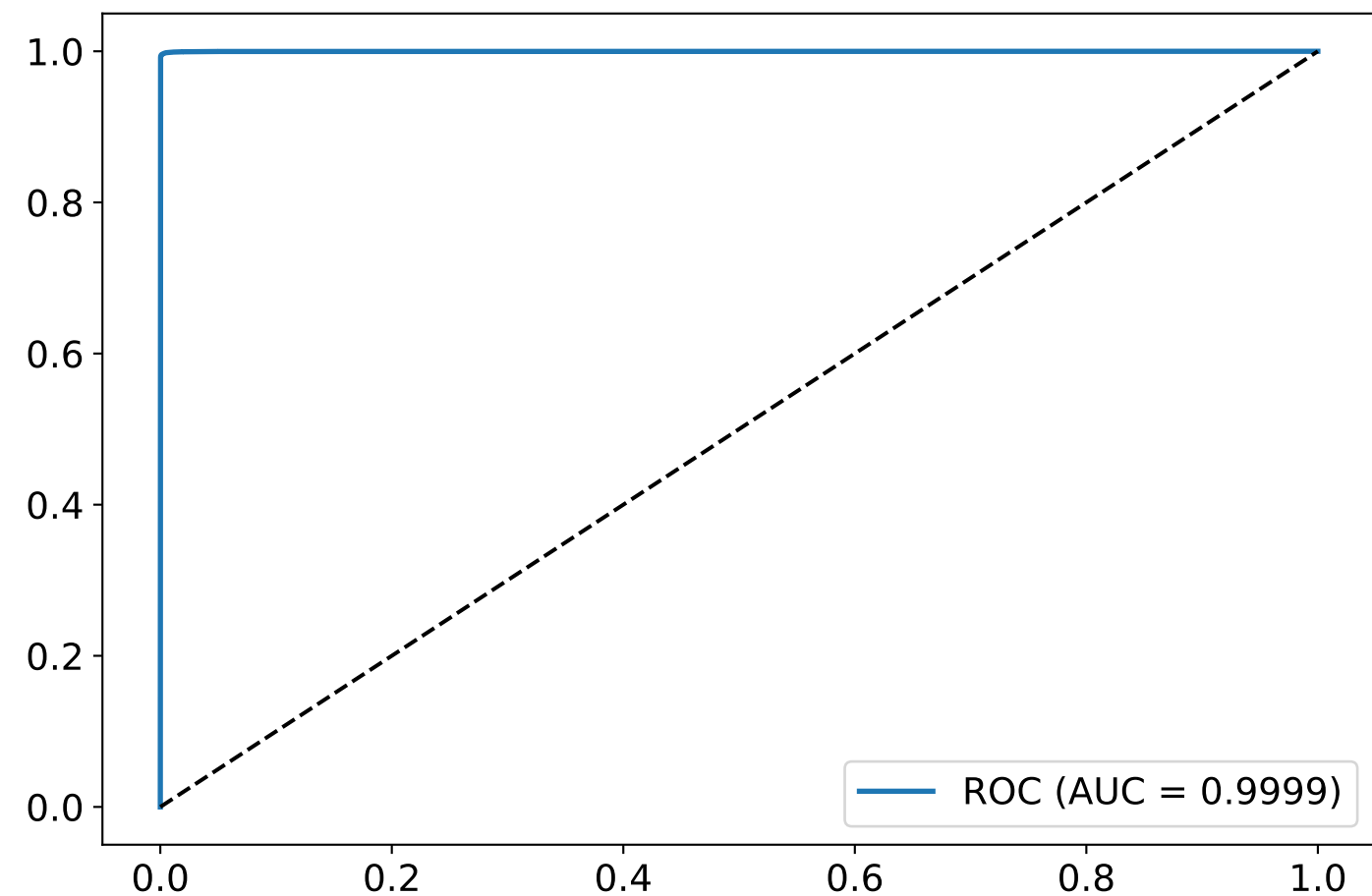


EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.997

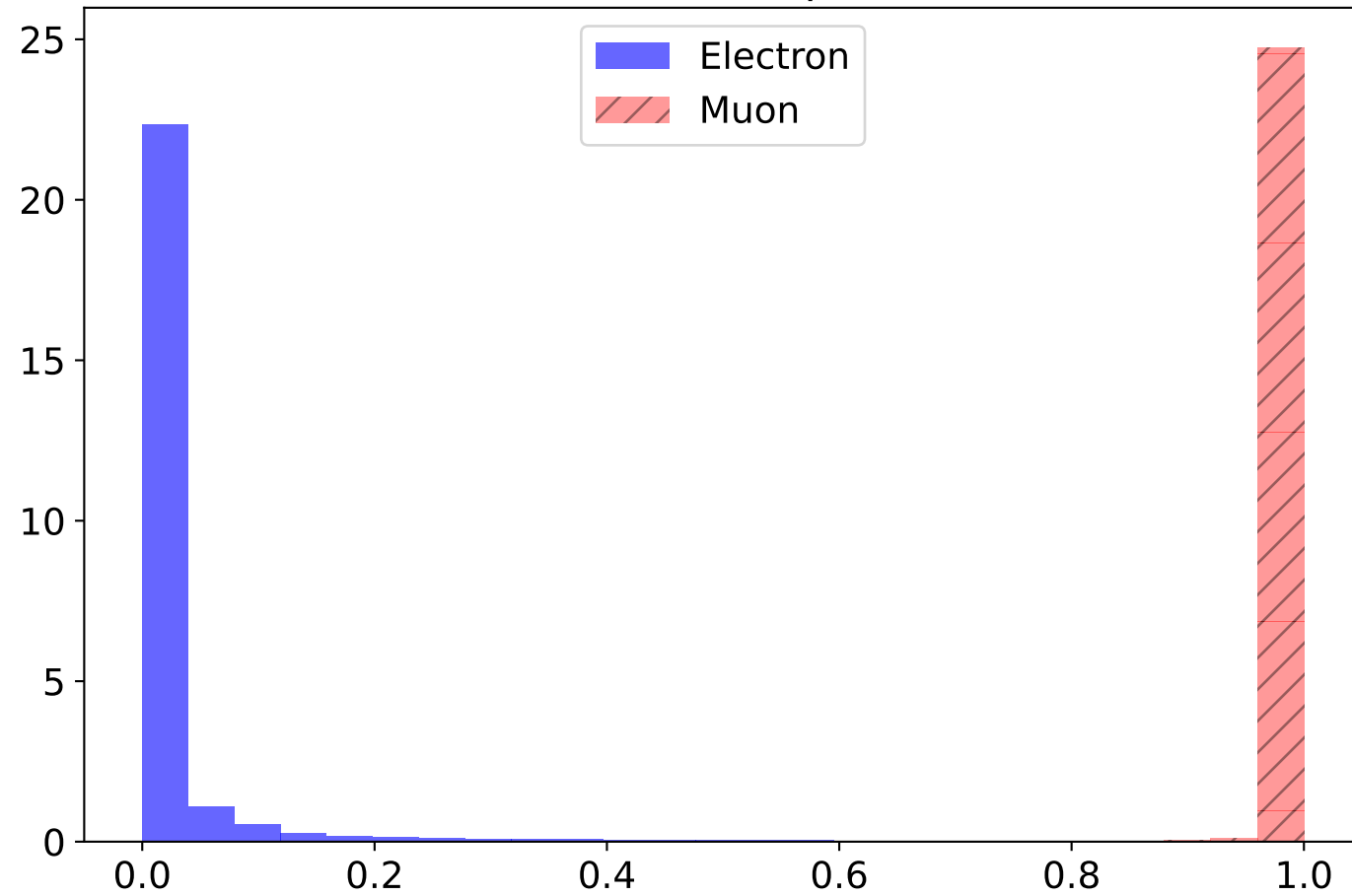
Confusion Matrix



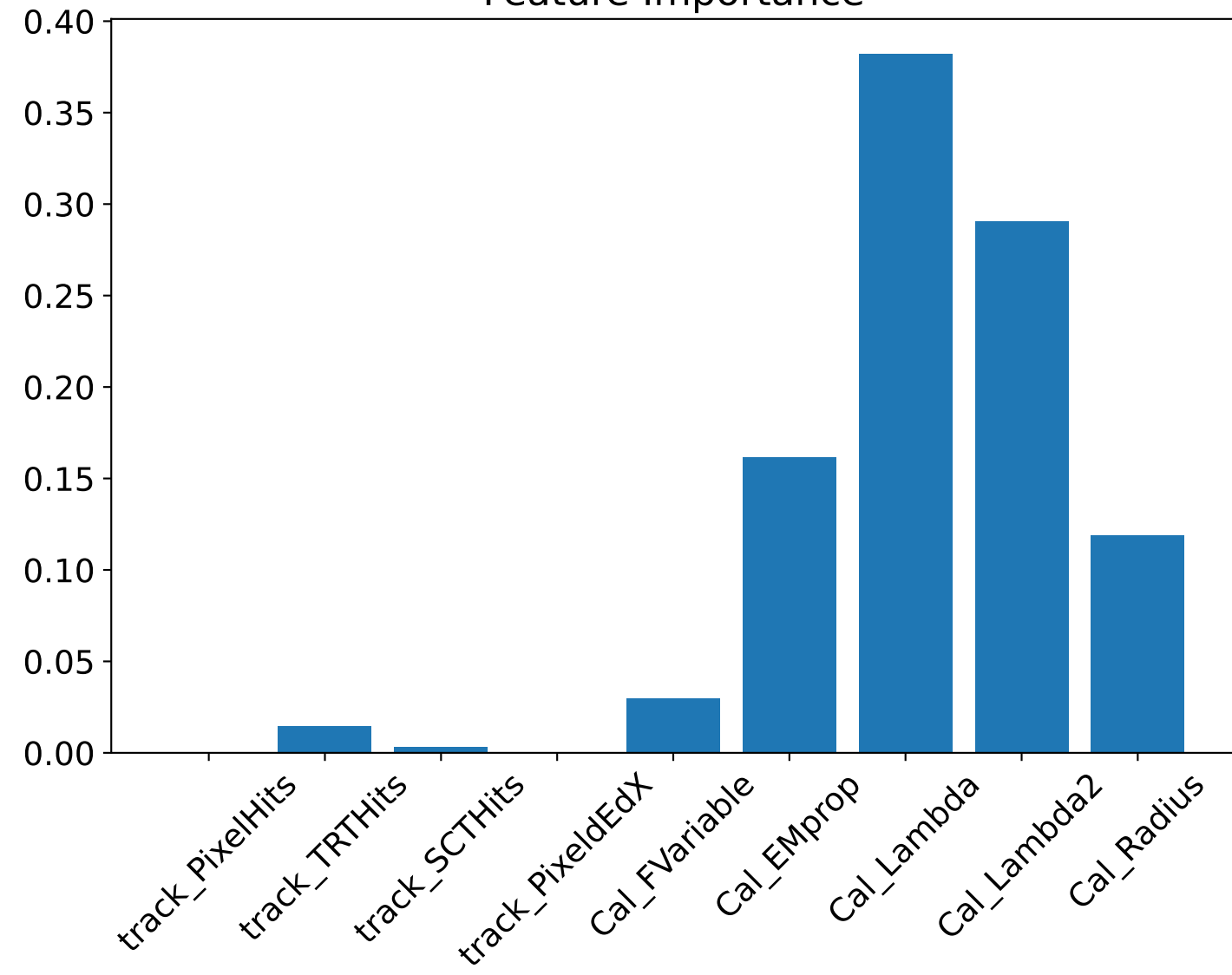
ROC Curve



Classifier Response

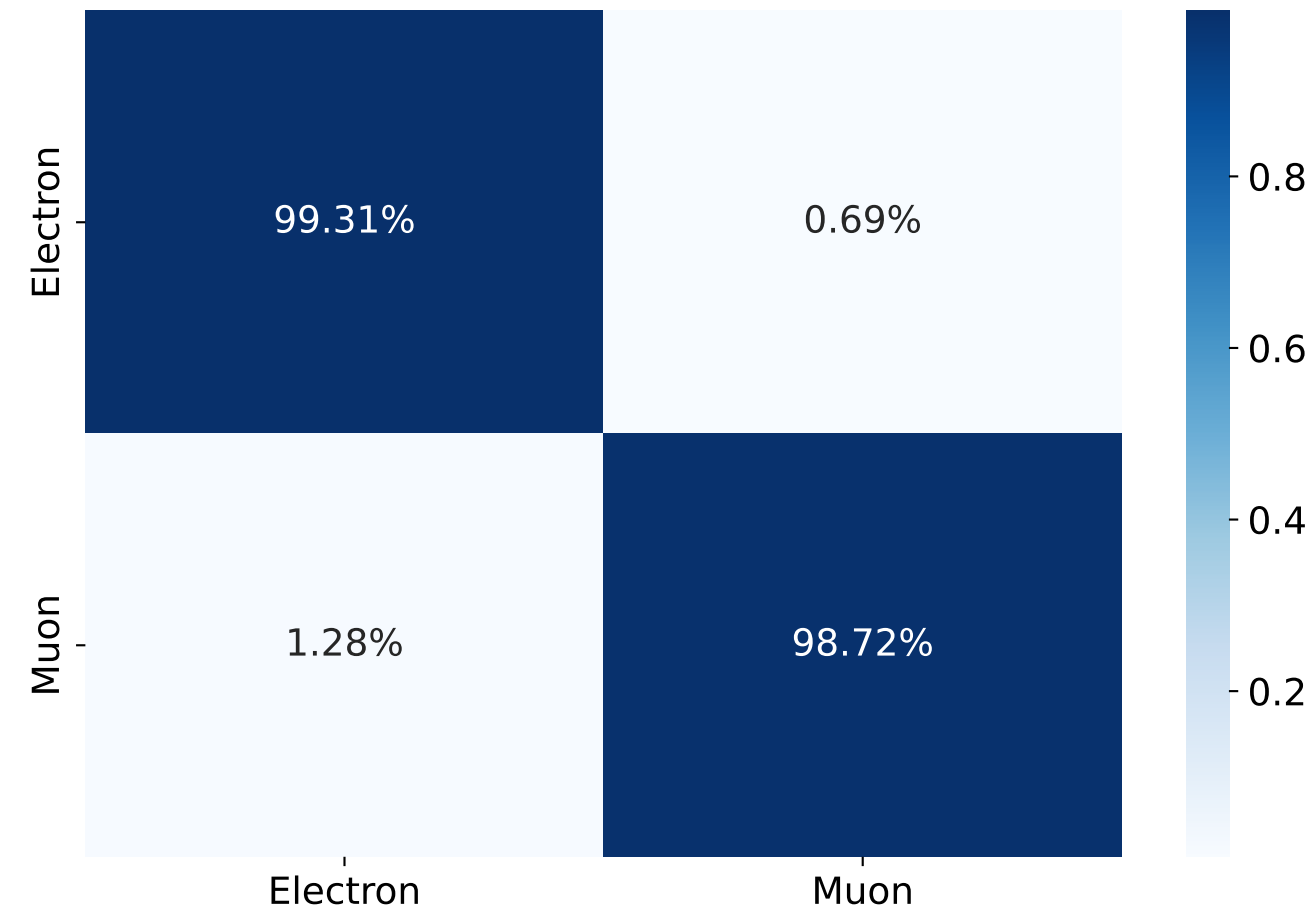


Feature Importance

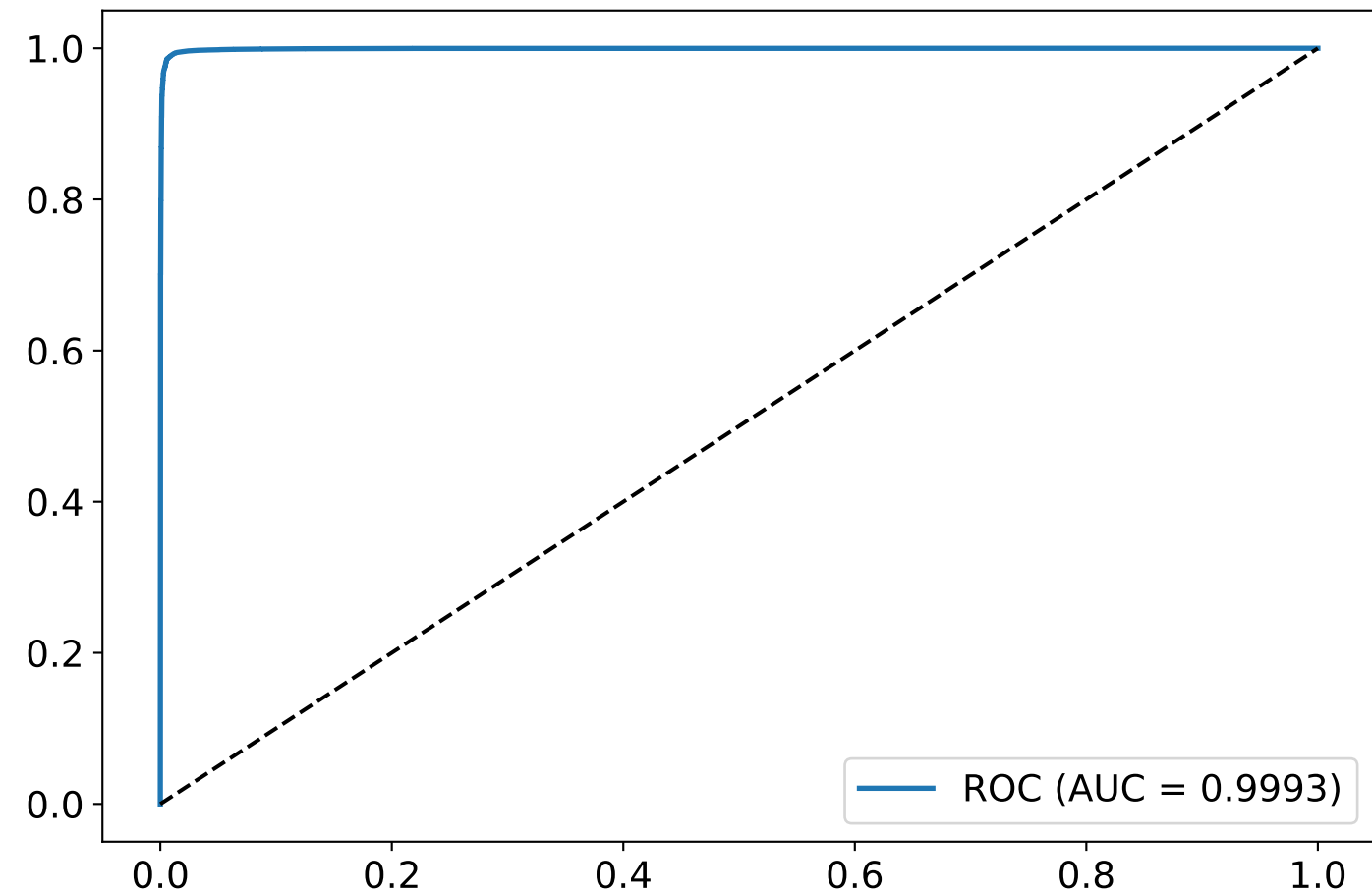


EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.988

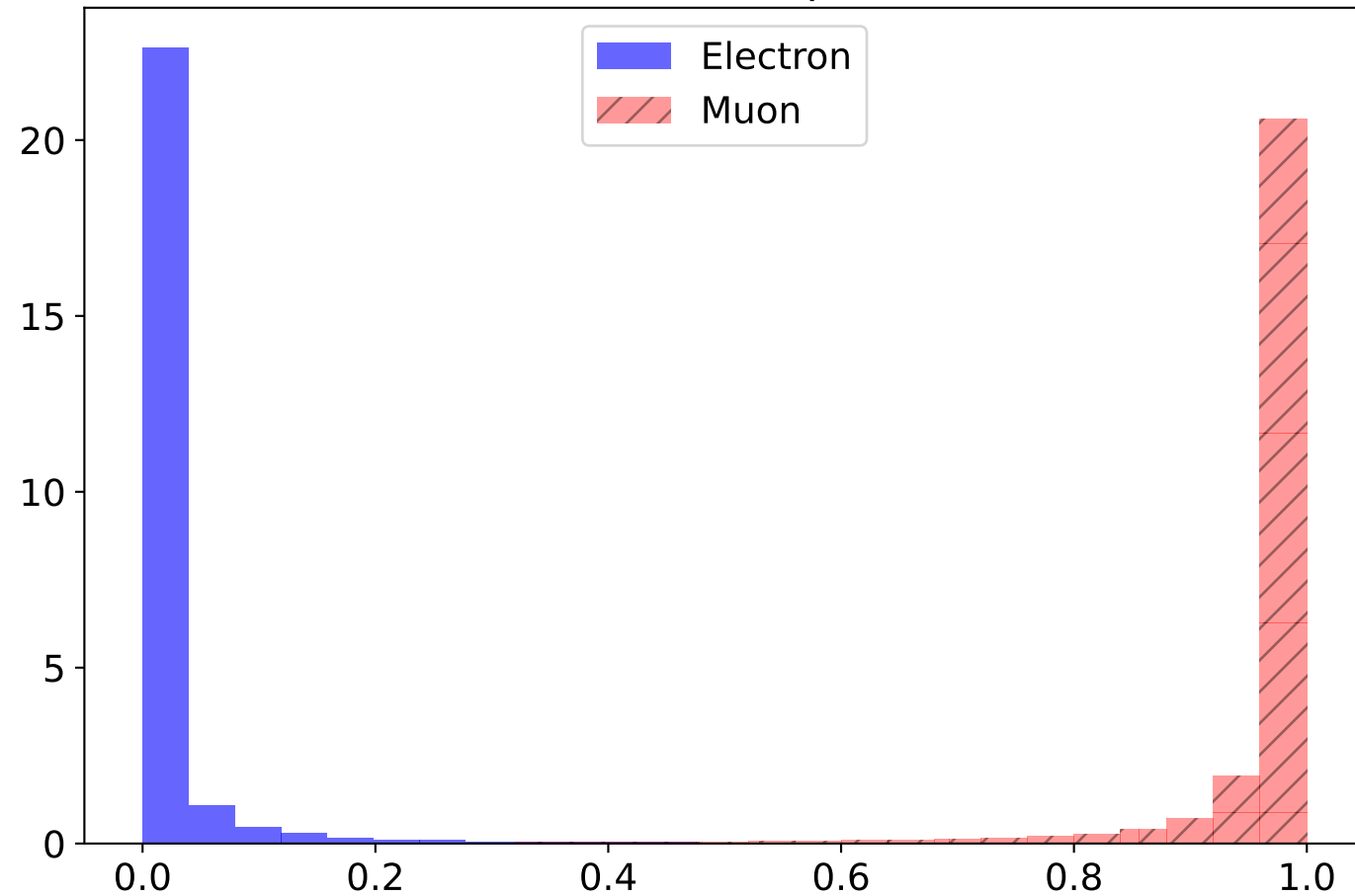
Confusion Matrix



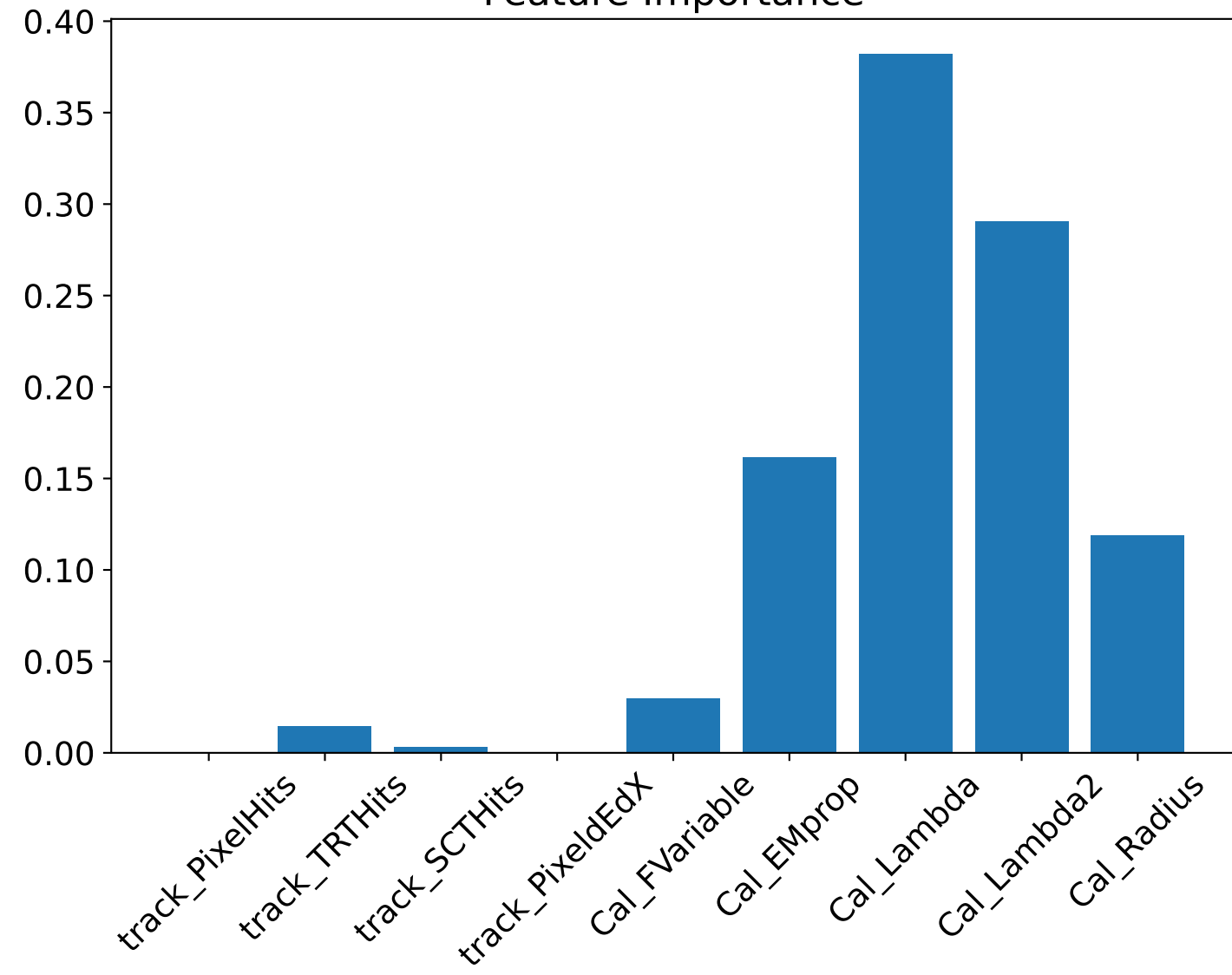
ROC Curve

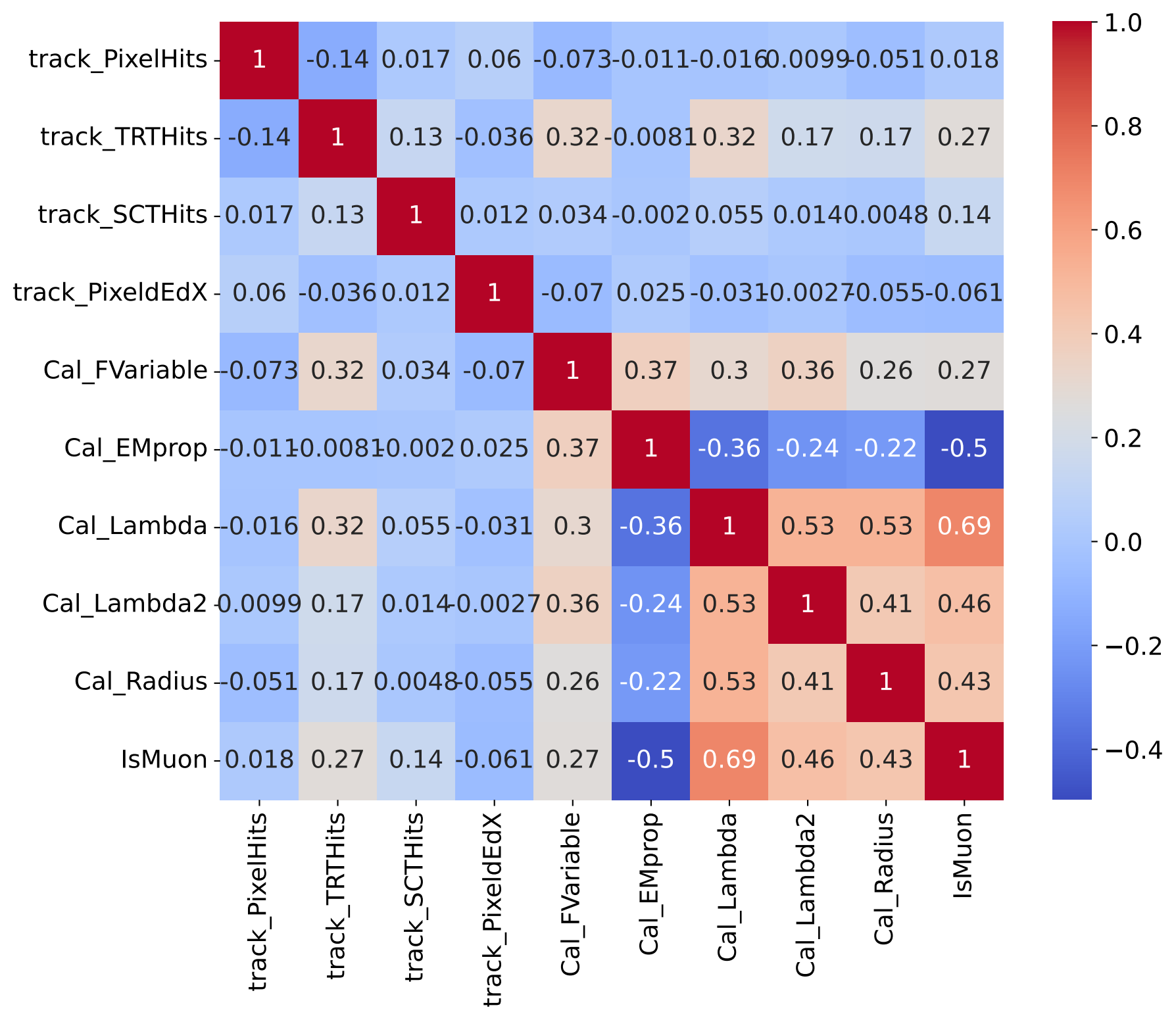


Classifier Response



Feature Importance





Classification Reports Summary

Trained: MCData.root | Test: MCData.root

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.98	1.00	0.99	104123
Muon	1.00	0.99	1.00	287040
accuracy			0.99	391163
macro avg	0.99	0.99	0.99	391163
weighted avg	0.99	0.99	0.99	391163

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.98	1.00	0.99	69342
Muon	1.00	0.99	1.00	160476
accuracy			0.99	229818
macro avg	0.99	0.99	0.99	229818
weighted avg	0.99	0.99	0.99	229818

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.99	0.99	20737
Muon	1.00	1.00	1.00	76139
accuracy			1.00	96876
macro avg	1.00	0.99	1.00	96876
weighted avg	1.00	1.00	1.00	96876

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.96	0.99	0.97	14044
Muon	1.00	0.99	0.99	50425
accuracy			0.99	64469
macro avg	0.98	0.99	0.98	64469
weighted avg	0.99	0.99	0.99	64469